Yeast 001 — the supercut, as cut

6:51 · 210 picks · 23.976p · rough cut at `build/roughcut/yeast001s.roughcut.fcpxml`

Both shoots. The June 16 reshoot (`footage/a-cam/`) is the spine; the June 9 tapes (`Intro 001/`) supply 1:38 of the lines that were simply better said the first time — the jar run, the "it's in mead, it's in wine, it's in beer" list, the close. Same table, same room; different top and hair, so the intercut reads as a second day, not a second location. Every source is level-matched to the cut's median in the 200 Hz–4 kHz band (from –2.5 dB on MVI_0019 to +4.3 dB on MVI_0037), and all of it takes the `yeast001-grade.cube` LUT.

Every explanation is the shortest take that still says the thing. The gooseneck experiment is kept whole. What's gone: the hazy-DNA aside, the spontaneous-generation preamble, the sourdough aside, mother-ferment → measured yeast, the glutathione mechanism, amylase and invertase, the proofing-temperature sentence, pre-ferment flavour. Each is listed at the end with where it lives if wanted back.

Pace. Every gap between two words longer than 0.4 s was measured off the audio: the silent part is kept as a 150 ms breath next to the next word, the voiced part — the "um" WhisperX never writes down — is cut. 210 picks for 6:51 is a cut every two seconds; that is the energy, and it is also why the jump cuts inside a take are covered by b-roll wherever there was any.

Pictures. The three Wikipedia pictures no longer sit over her. When one comes in, the footage snaps to half size on the right inside a paper mat and the picture takes the left at full height; when it goes, the footage snaps back. Pictures are dead still — no jitter, no rotation, hard in, hard out. The chapter cards still step and settle per §5.

The read-back is in `read-back (what the cut actually says).txt` : two flags, both her own parallel phrasing, not doubled takes.

Cut	6:51 — 15% of the 52 minutes shot across both days
Music	*Twin Lakes* under the cold open, out by 0:30. *Two Big Reasons* from 6:19, ending on the last frame.
Pushes	1.05× under the hook claim (0:05), 1.06× under "it's really all just the same" (3:11), 1.05× under "you do not actually need to add sugar" (6:10), 1.04× under the close. No static punch-ins — there is no take long enough to need one.
Grade	LUT on every clip; face lift (preview only) — build sheet §G in `cuts/yeast001/` .

Cold open — 0:00

> This is yeast bubbling in action. And this is me, way too excited about yeast.
>
> You don't actually need sugar to make your yeast grow. You just need time.
>
> Hi, I'm Katie. I'm a food scientist, and this is my deep dive into yeast.

1. Yeast is old — 0:13

> Yeast is really old. We've had it around for at least a few thousand years. We've found it in archaeological records in Egypt. We have found it in archaeological records in parts of Asia and in Europe.
>
> *(June 9)* It's being passed between brewers and bakers. It's being collected from the air. It's being collected from outside of grapes and introduced into wine. It's moving around. It's passed from person to person, baker to brewer, brewer to baker. It's in mead. It's in wine. It's in beer. It's in our sourdough starters. It's just all around.
>
> They may not know what they're breeding, but they're being selective about it. If it's a dud batch, they're not passing it on to their friend.

> *(June 9)* And so what we're getting is selective breeding of good yeast — yeast that ferments, that makes the bread bubble, makes the alcohol happen — is being passed around. And that is what we've used for years and years and years.

⚠ "It's in mead" transcribes as "meat" every time. Captions, not audio.

2. Louis Pasteur — 1:02

> It wasn't until we meet Louis Pasteur that we have an inkling of what's making one successful batch work, and what's being passed on between these bakers and brewers — has a gooseneck experiment, and disproves spontaneous generation, and says there are microbes, little bugs that we can't see, all around the world.
>
> If you don't know about the Louis Pasteur experiment: what he does is he takes a glass bulb and stretches out a neck on it, that is open over here, but it has a gooseneck shape. And he heats it up to kill off any bacteria that's on the inside, with some food that he has cooked, that he's put inside of this gooseneck bottle. And so if there's any truth to spontaneous generation, the food will just break down. There will be no problem with it doing what food has always done. But if the food breaks down because of germ theory — what he was thinking: germs can't get in through the gooseneck bottle. It falls down. It gets stuck in the downspout, right here. So nothing will happen to the food. It won't break down. It won't do anything. And that's what happened. And so Louis Pasteur proves that there is no spontaneous generation — that there is something in the air: bacteria, yeast, living organisms that are getting into our food and making changes.

⚠ "brewers — has a gooseneck experiment": the "He" is gone because the take says "He uses— has", and WhisperX had folded "uses" into the "He" token, so there was no clean "He" to keep. One word in a pickup if it bothers you. The Pasteur portrait reframes the footage at 1:04 for 7 s; the flask at 1:23 for 12 s.

3. One bug — 2:13

> We meet our *Saccharomyces cerevisiae*. And this is the most common form of yeast that we use today. There are other yeasts that are used in sourdoughs, but for the most part, when you get to industrial bakers and you get to home bakers and you get to brewers, they're all using the same little bug: *Saccharomyces cerevisiae*.
>
> *(June 9)* It is like 90% of all the yeast that we use is this one bug. All over — we use it everywhere, from our home bakeries to the brewers down the road to the vineyards in California and France. Everywhere. *Saccharomyces cerevisiae*. Same guy, again and again and again.

⚠ 90% goes on screen as a number (§6.8 wants a source). The micrograph reframes at 2:19 for 8 s.

4. What's on the shelf — 2:51

> *(June 9)* I have marketing for yeast for mead. It's *Saccharomyces cerevisiae*. I have instant dry yeast. It's *Saccharomyces cerevisiae*. I have active dry yeast with a little sugar in it. Guess what? *Saccharomyces cerevisiae*. Fresh yeast. *Saccharomyces cerevisiae*. I even have what could be the start of a sourdough starter. Here, it's a pre-ferment. It is yeast, *Saccharomyces cerevisiae*, mixed with flour and water.
>
> And so someone with a marketing degree, with a business degree, is saying: hey, let me sell to you. I know what you need. It's this. But it's really all just the same *Saccharomyces cerevisiae*.

The jar run is cut on each item — mead packet, fine granules, the dry jar, the fresh cube, the tall glass — from the June 16 b-roll, over the June 9 sound.

5. Access and need — 3:23

> Why do we have so many different names for it? Why do we have so many different formats for it? The answer is access and need.
>
> You have the at-home baker in the middle of a farm in Kansas that needs something every once in a while, and doesn't want to have to drive two hours back and forth to a grocery store every couple days to get fresh yeast. They need something on hand that they can keep, that won't die off on them as frequently. So you have fresh yeast and dry yeast, and you have active dry and instant.
>
> But in the processing of the active dry yeast and the instant yeast, the yeast gets a little damaged. It helps dry out the bread a little bit faster. And to be honest, an at-home baker might not care. You might not notice. They consume the bread within a couple days, because they're only making one loaf at a time. Because it might not be worth driving a couple hours to a grocery store to get something to have a bread that's a little bit stretchier that lasts a day longer.

The glutathione mechanism is out (it was escalated anyway — E3). "Gets a little damaged — it helps dry out the bread a little bit faster" is the whole claim now.

6. Budding — 4:15

> So you have a mother cell that grows, and buds on itself a daughter cell. And when that daughter cell is big and identical to the mother cell, it uses mitosis and splits into two cells. The yeast will continue to grow and reproduce itself when it's in an environment that has lots of nutrients for it. And those nutrients are sugars and starches. It breaks down starch into sugar.

7. What it eats — 4:36

> The yeast will use simple sugars and break it down with the enzyme zymase into CO_2 and alcohols. And the CO_2 and the alcohol are the byproducts that we're looking for as humans — either in alcohol production, or in helping our bread rise with the CO_2, giving the gas that we will eventually bake off.
>
> *(June 9)* And you also have your CO_2. That's where you get your bubbly wines, like your champagne.

Amylase and invertase are out. Sugar → zymase → CO_2 + ethanol is the one line left, and it is still E2 until she signs it.

8. Temperature — 5:00

> Yeast is looking for a right environment for it to thrive and to grow, and that environment has lots of sugars and starches, and is in a temperature between 50 degrees, where it kind of shuts down and goes dormant, and 140 degrees, when it's completely killed off. There's a nice sweet spot right in the middle — about 93 degrees Fahrenheit and 100 plus a couple degrees Fahrenheit. And in that sweet temperature spot, it will grow and thrive.

⚠ E1 — the numbers. "Fifty" is a quarter-second blip inside a pause the transcript mistimed; the in-point is set off the audio so it survives.

9. Faster and slower — 5:29

> So when we add salt and fat and other spices to our bread doughs, what we're doing is we're putting items in the way of the yeast and its food. And so it's slowing down the process in which the yeast can digest the sugars that it wants for growing, and that will then slow our proofing time. The availability of sugar will speed up the proofing time.
>
> This is why in active dry yeast they recommend that you add a little bit of sugar, to kind of start the yeast up and going and getting it bubbling. You don't have to actually do that. You can just let it sit a little longer. Yeast itself just really wants

> flour and water. You can do that sometimes by getting a pre-ferment. This is what I have going right here, and it's already starting to bubble up, and it's just flour and water and yeast, and the yeast is in a happy spot.
>
> Adding sugar does speed it up, but then you end up with a sweeter dough. And if that is not the final product that you want, you do not actually need to add sugar to your dough to get the yeast to grow.

10. The same little bug — 6:26

> *(June 9)* Use what's available. It's all the same. It can be played with. Yeast is not something that has to be a hard and fast rule. It feels like a mystery, but really all it is is the same little *Saccharomyces cerevisiae* bug, again and again and again and again.
>
> All the little bug wants is to go and eat sugars and fart — produce carbon dioxide and ethanol. It just wants to live a happy life, and it's not all that complicated.

⚠ "…and fart" is in. It is the best line on either tape and it is also a decision about what the channel sounds like; the cut straight from "eat sugars" to "produce carbon dioxide" is clean if it goes. ⚠ Still no next action (D1).

What was cut, and where it lives

Line	Where	Why
"The details are a little hazy… hard to extract the DNA"	MVI_0036 @ 1:02	slow, first forty seconds
"So it's natural that ancient people would have just stumbled upon it" / the yeast-in-the-air passage	MVI_0037 @ 0:55, MVI_0038 @ 3:32	the June 9 list says it faster
"Now, an aside about sourdough…"	MVI_0042	an aside
mother ferment → measured yeast; "our way of baking changes"	MVI_0038 @ 9:07–10:05	explanation, not argument
industrial bakers don't want to reactivate; "why doesn't the large baker use instant"	MVI_0040 @ 0:47, 2:03	the Kansas half carries it
glutathione, the gluten network, sandwich bread	MVI_0040 @ 2:52–3:23	escalated (E3), and a mechanism
"how do we reproduce it?"	MVI_0040 @ 4:37	the answer is enough
amylases, sucrose and invertase	MVI_0040 @ 6:28–6:57	escalated (E2); zymase is the line
"that is the temperature we generally want our bread to be sitting at"	MVI_0040 @ 8:51	said by the gauge

Line	Where	Why
the pre-ferment's more complex flavour	MVI_0040 @ 11:40	tail

The long cut (`cuts/yeast001/` , 13:04) keeps all of these and is the one to go back to if a chapter needs its full length.